CIP-B101
LABORATORY REPORT

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

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Registration Number	
Course Code	CIP-B101
Course Title	Basic Computer Skills for Digital Forensics
Lab Title	Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis
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Platform Used	Windows 10 Home (Kali Linux used for the sector check in C1)
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1. Introduction

In this lab I looked at my own laptop the way a forensic investigator would. I found out what the computer is, what is on its disk, and how the disk is divided into partitions. I also calculated the size of two partitions from their sector counts and compared my answers with what Windows reports.

I only looked and calculated. I did not format, delete, resize or change anything on the disk.

2. Tools Used

- System Information (msinfo32) and Command Prompt: system details, disk details and running processes
- Disk Management, PowerShell and DiskPart: the partitions on my disk (listing and viewing only)
- A calculator: all size calculations
- Kali Linux terminal: to check the slide calculation in C1

Safety: I used only read-only commands. In DiskPart, I only used list disk, select disk 0, list partition and exit. I covered personal details such as my computer name, username and serial number in the screenshots.

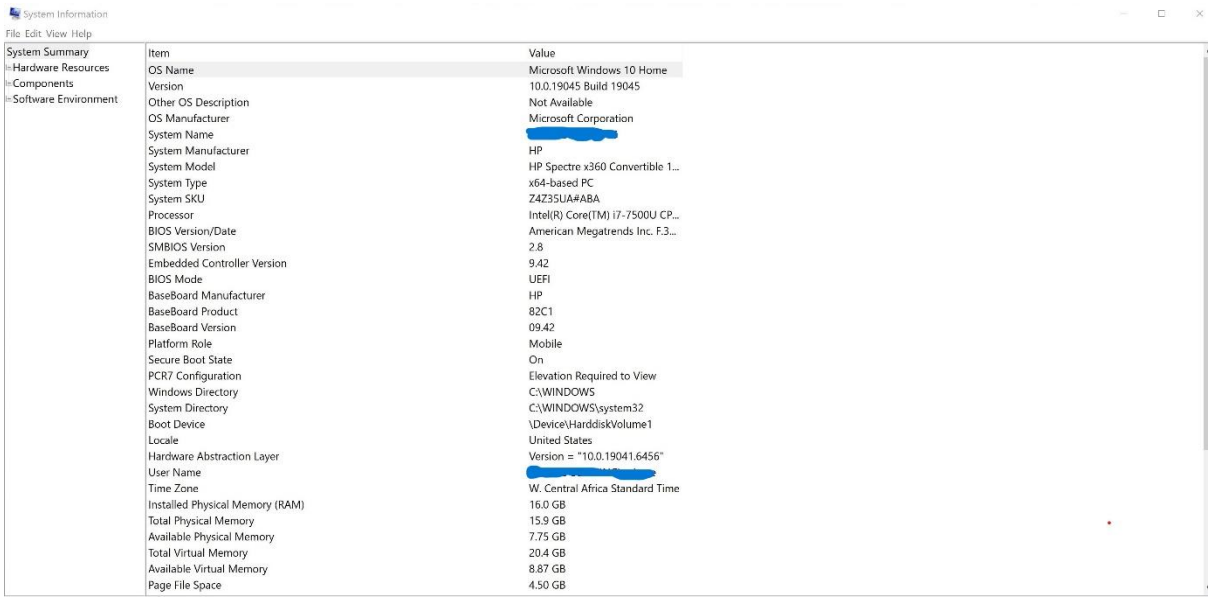
Part A - System Identification & Evidence Layer Mapping

A1. System specification

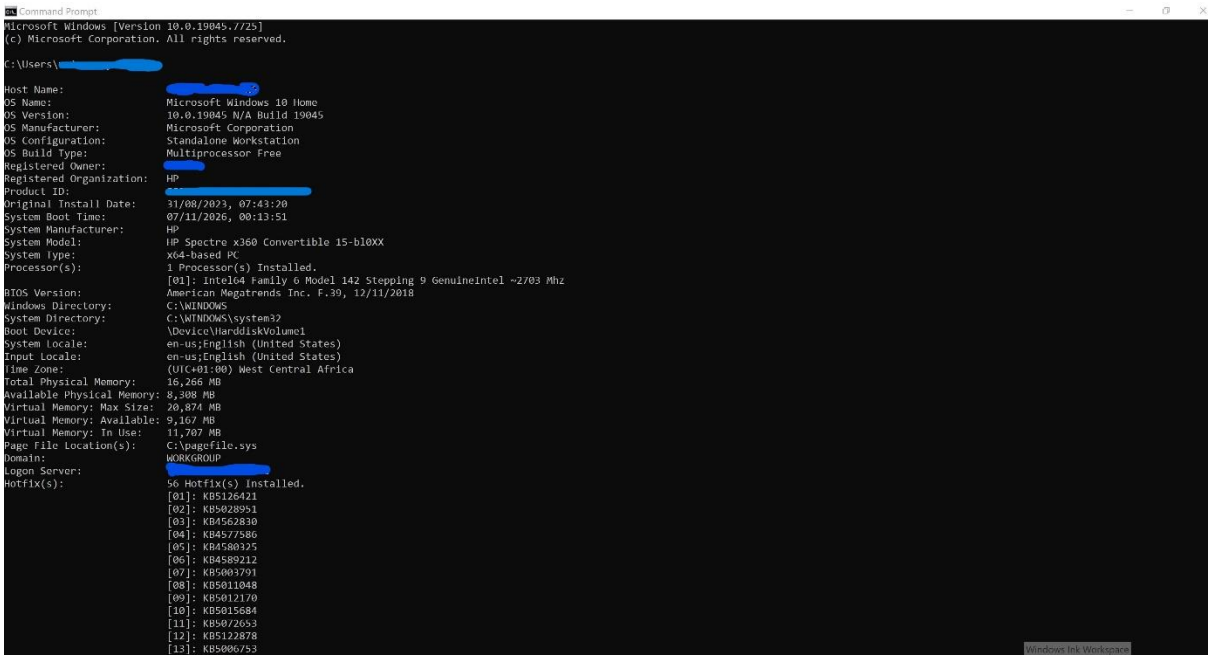
I opened System Information (msinfo32) and also ran the systeminfo command.

```
msinfo32
systeminfo
```

Item	What it showed
Operating system	Microsoft Windows 10 Home, Version 10.0.19045 Build 19045
Computer	HP Spectre x360 Convertible 15-bl0XX
System type	x64-based PC
Processor	Intel Core i7-7500U
BIOS Mode	UEFI
Secure Boot State	On
Memory (RAM)	16.0 GB installed (15.9 GB usable)



Screenshot 1a: System information using msinfo32



Screenshot 1b: System information using systeminfo in command prompt

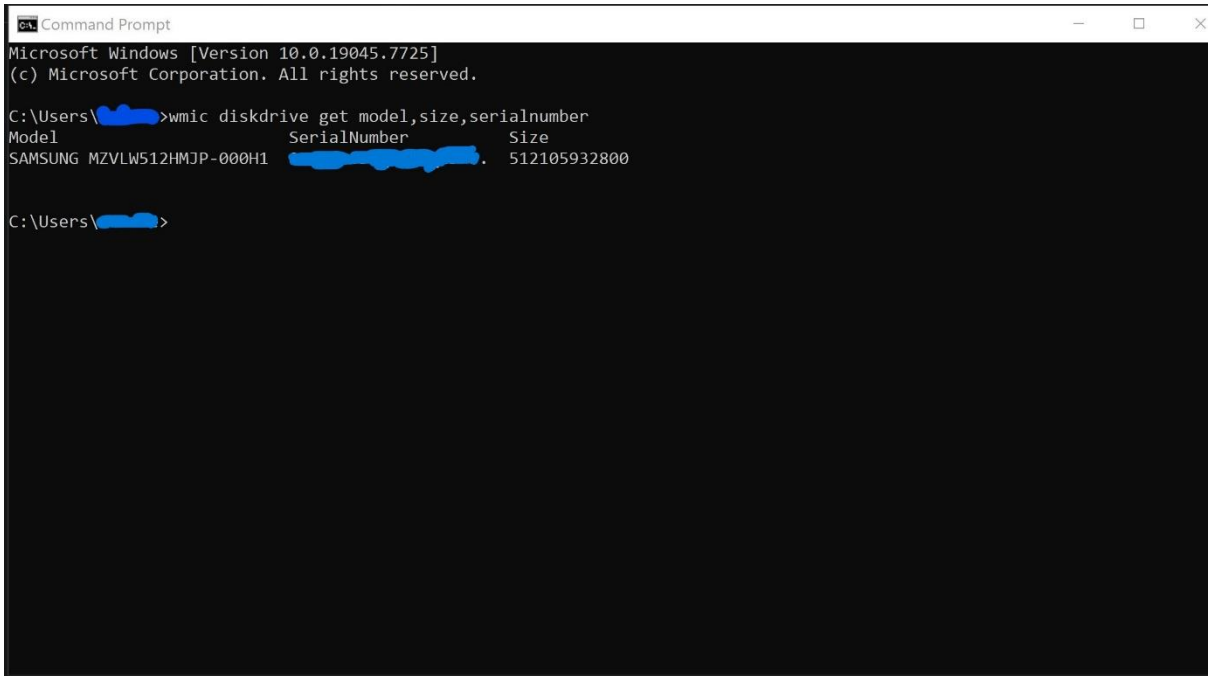
What I found: both tools gave the same operating system, model and time zone. My laptop starts in UEFI mode with Secure Boot turned on.

A2. Storage information

```
wmic diskdrive get model,size,serialnumber
wmic logicaldisk get caption,filesystem,size,freespace,volumename
```

The physical disk is a SAMSUNG MZVLW512HMJP-000H1. Its size is 512,105,932,800 bytes.

Drive	Name	File system	Size in bytes
C:	Windows	NTFS	498,841,153,536
D:	RECOVERY	NTFS	11,945,373,696
E:	no details shown	no details shown	no details shown



Screenshot 2: Disk model and size



Screenshot 3: Logical volumes / file systems

What I found: the disk is about 477 GB when divided by 1024 three times (the disk is sold as 512 GB). Drive E: showed no file system or size, so I could not tell what it is.

A3. Evidence layers

Different parts of a computer keep different evidence. The idea is: user activities = applications + operating system + disk/memory.

Layer	Example evidence	Why it matters
User	Recent files, browser history	Shows what the person did
Application	Browser cache, document details	Shows which program was used
Operating system	Event logs, user accounts	Shows system and account activity

Layer	Example evidence	Why it matters
Disk	Deleted files, partitions	Helps recover files and build a timeline
Memory / process	Running programs and their PIDs	Shows what was happening at that moment

The same action can leave traces in more than one layer. For example, opening a document can appear in the recent files list, in the operating system log, on the disk and in memory. If one trace is deleted, another may remain, so an investigator checks all the layers.

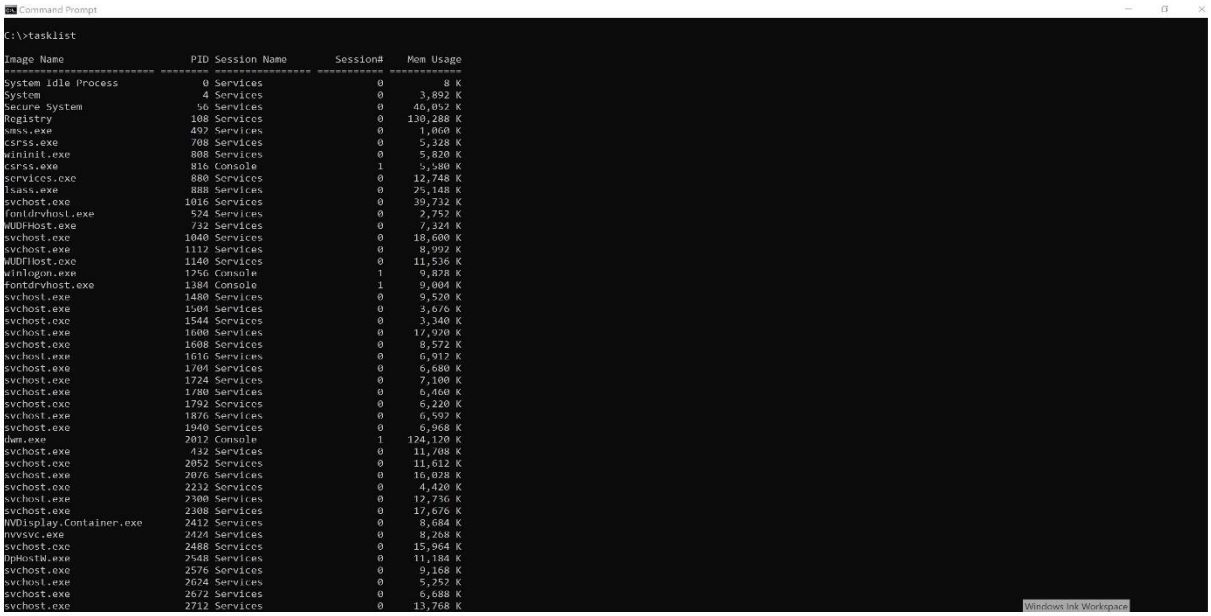
This makes computer forensics technically difficult because computers have different operating systems, applications, hardware components and connected devices that generate different types of digital evidence. The large volume and different formats of data also make evidence identification, collection and interpretation challenging.

Part B - Program, Process and Thread Investigation

B1. Observe running processes (three processes)

```
tasklist
```

Process	PID	What it does
explorer.exe	10480	Runs the desktop, taskbar and File Explorer
cmd.exe	16528	The Command Prompt window I used for the lab
chrome.exe	2924	The Google Chrome browser (started by me)



Screenshot 4: Running processes

PIDs change every time a program starts, so these numbers are only valid for the time I took them.

B2. Program, process and thread

- A **program** is a file on the disk with instructions, like chrome.exe.
- A **process** is a program that is running now, and it has a PID.
- A **thread** is a smaller job inside a process; one process can have many threads at the same time.

Example: a recipe book is the program, cooking in the kitchen is the process, and the cooks working together are the threads.

Why it matters: a process list shows what was running on a computer at one moment. Investigators use it to spot unknown or suspicious programs. It is lost when the computer is switched off, so it should be recorded early.

Part C - Disk Sector and Disk Size Calculation

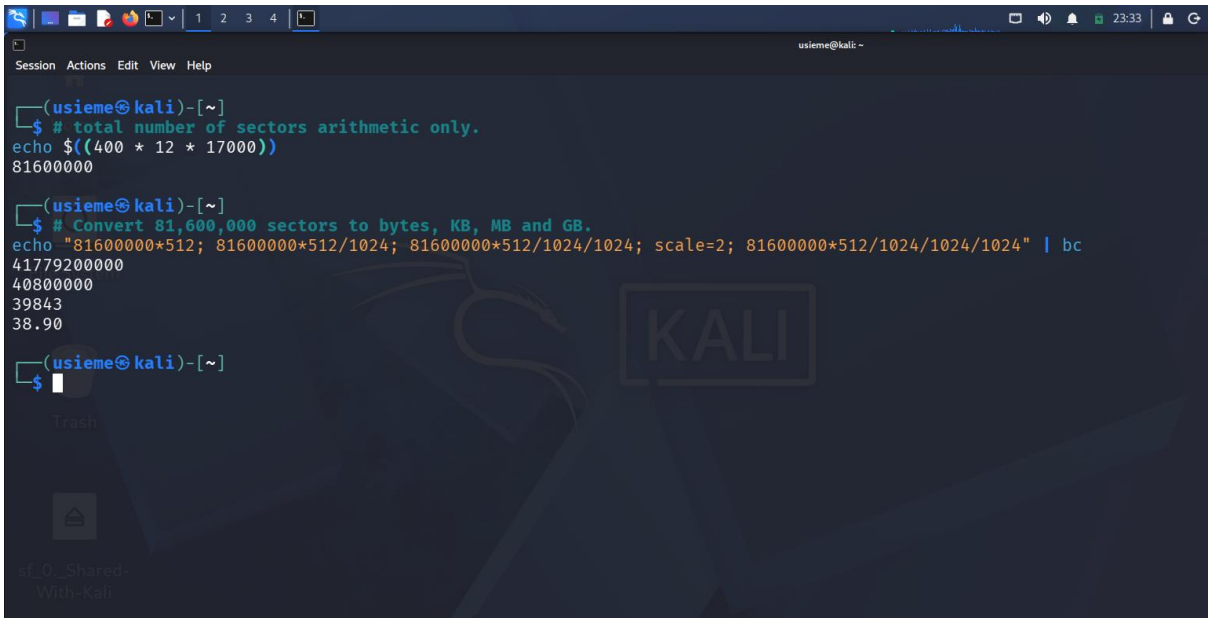
Rule: size = number of sectors x bytes per sector. To change bytes to GB, divide by 1024 three times.

C1. Slide example

Given: 400 sectors per track, 12 heads, 17000 cylinders, 512 bytes per sector.

```
Sectors = 400 x 12 x 17000      = 81,600,000
Bytes   = 81,600,000 x 512      = 41,779,200,000
KB      = 41,779,200,000 / 1024 = 40,800,000
MB      = 40,800,000 / 1024     = 39,843.75
GB      = 39,843.75 / 1024      = 38.91
```

I worked this out with a calculator and checked the sector count in Kali Linux with echo $((400 * 12 * 17000)), which gave 81600000. In my screenshot the bc tool shows 39843 MB and 38.90 GB, because it cuts off decimals instead of rounding. The calculator answers are the correct ones.



Screenshot 5: Total number of sector calculation

C2. My own partitions

The offsets and sizes below come from the Get-Partition output, which is shown in Part D (section 6). Sector size = 512 bytes. Start sector = offset / 512. Sectors = size / 512. End sector = start + sectors - 1.

Partition 1 (EFI System Partition)

```
Start sector = 1,048,576 / 512     = 2,048
Sectors      = 272,629,760 / 512   = 532,480
End sector   = 2,048 + 532,480 - 1 = 534,527
```

```
Size in GB   = 532,480 x 512 / 1024 / 1024 / 1024
             = 0.2539 GB (about 260 MB)
```

Partition 2 (Reserved)

```
Start sector = 273,678,336 / 512   = 534,528
Sectors      = 16,777,216 / 512    = 32,768
End sector   = 534,528 + 32,768 - 1 = 567,295
Size in GB   = 32,768 x 512 / 1024 / 1024 / 1024
             = 0.0156 GB (16 MB)
```

Partition 3 (Windows, C:)

```
Start sector = 290,455,552 / 512           = 567,296
Sectors      = 498,841,157,632 / 512       = 974,299,136
End sector   = 567,296 + 974,299,136 - 1   = 974,866,431
Size in GB   = 974,299,136 x 512 / 1024 / 1024 / 1024
             = 464.58 GB
```

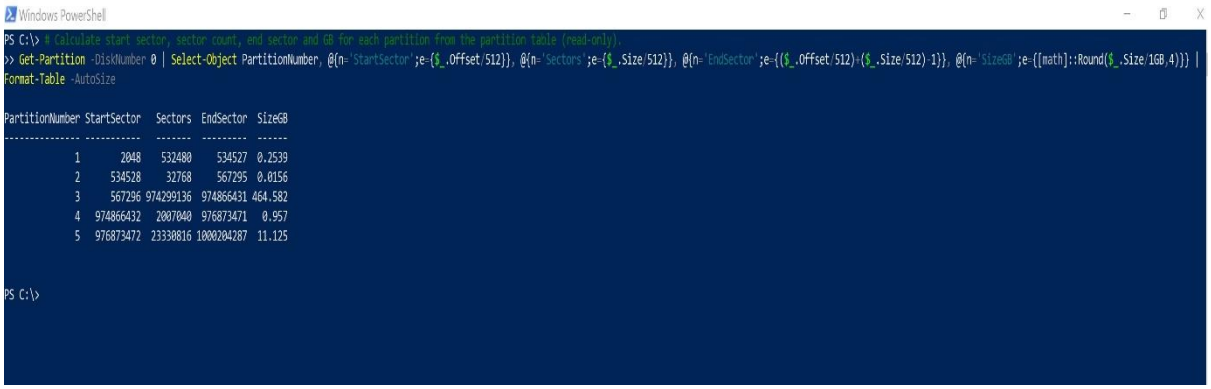
Partition 4 (Recovery)

```
Start sector = 499,131,613,184 / 512       = 974,866,432
Sectors      = 1,027,604,480 / 512         = 2,007,040
End sector   = 974,866,432 + 2,007,040 - 1 = 976,873,471
Size in GB   = 2,007,040 x 512 / 1024 / 1024 / 1024
             = 0.9570 GB (about 980 MB)
```

Partition 5 (RECOVERY, D:)

```
Start sector = 500,159,217,664 / 512       = 976,873,472
Sectors      = 11,945,377,792 / 512        = 23,330,816
End sector   = 976,873,472 + 23,330,816 - 1 = 1,000,204,287
Size in GB   = 23,330,816 x 512 / 1024 / 1024 / 1024
             = 11.125 GB
```

Each partition starts right after the previous one ends (for example, partition 1 ends at 534,527 and partition 2 starts at 534,528), so there are no gaps between the five partitions. I also checked all my answers with a PowerShell command (see the Appendix), and they matched.



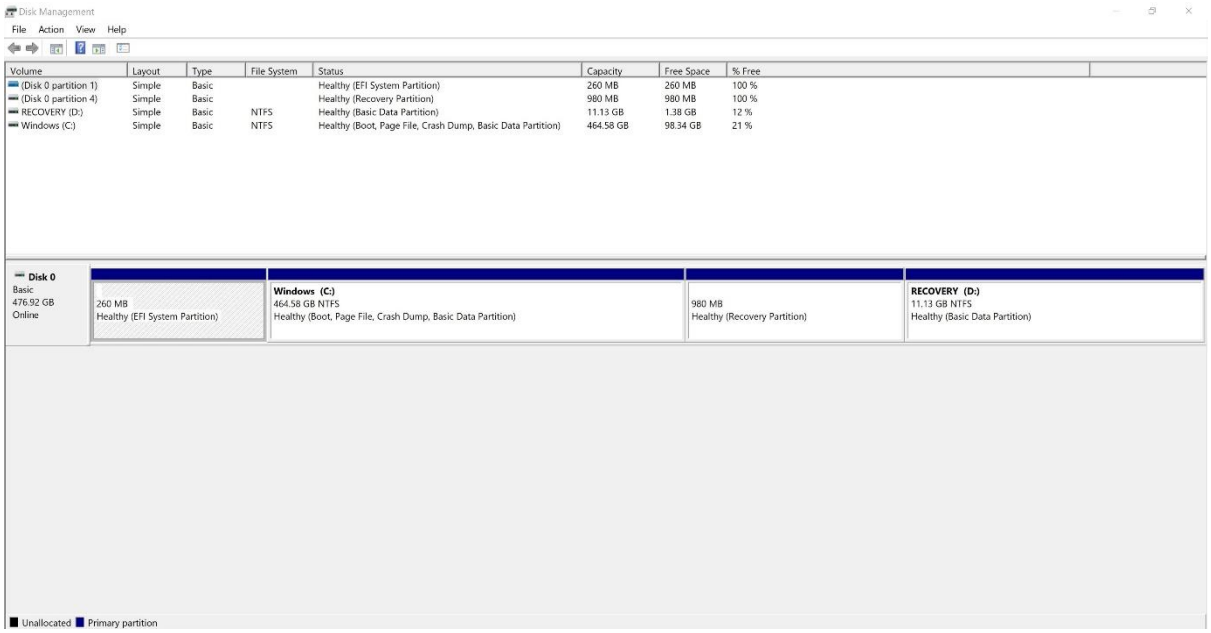
Screenshot 6: PowerShell calculation check for all five partitions

Part D - Safe Partition Table Inspection

A partition is a section of the disk. I used three Windows tools, all read-only, to see how my disk is divided.

D1 - Disk Management

```
diskmgmt.msc
```



Screenshot 7a: Disk layout and visible partitions using diskmgmt.msc in windows + R

Disk Management showed Disk 0 (476.92 GB) with an EFI System Partition (260 MB), Windows C: (464.58 GB, NTFS), a Recovery Partition (980 MB) and RECOVERY D: (11.13 GB, NTFS).

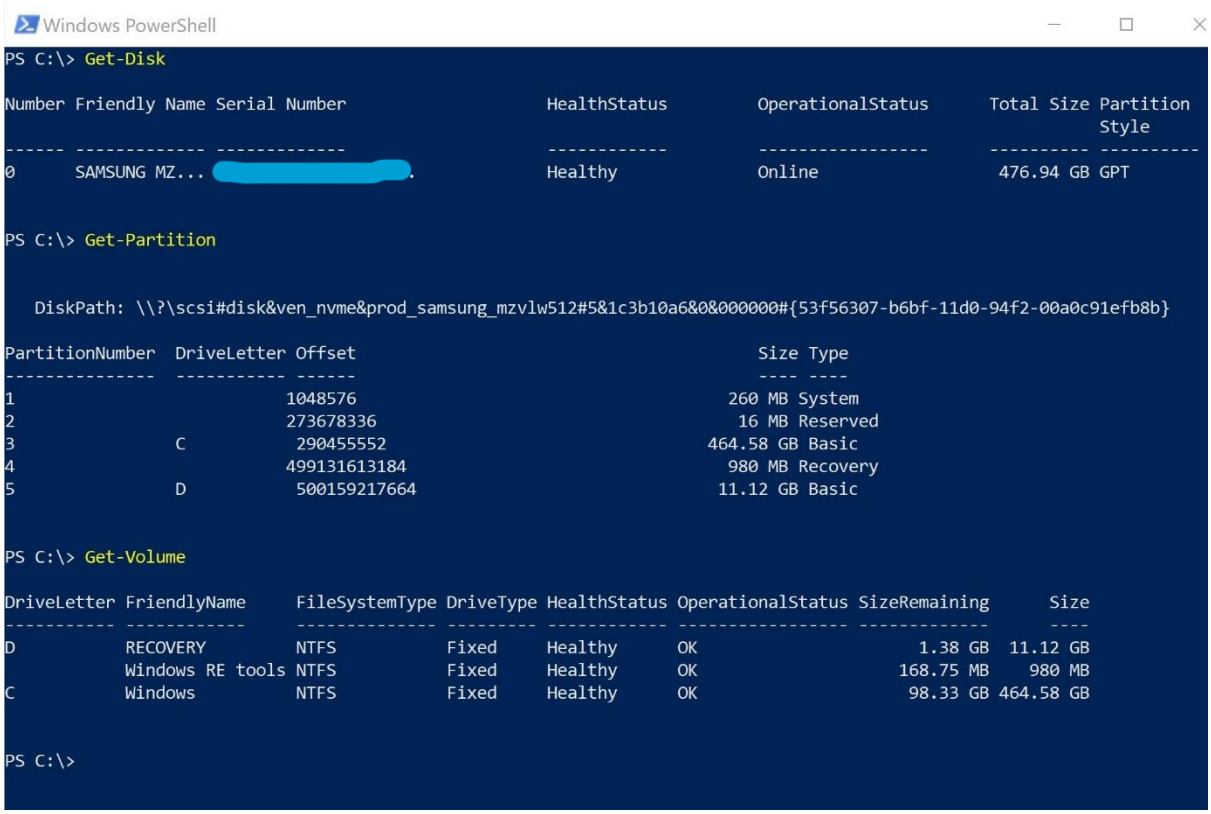
D2 - PowerShell

```
Get-Disk
Get-Partition
Get-Volume
Get-Disk | Select-Object Number, LogicalSectorSize, PhysicalSectorSize, PartitionStyle
Get-Partition -DiskNumber 0 | Select-Object PartitionNumber, DriveLetter, Offset, Size, Type | Format-Table -AutoSize
```

The disk uses the GPT partition style. Each sector is 512 bytes (logical sector size). Here are the five partitions:

Partition	Type	Offset (bytes)	Size (bytes)
1	System (EFI)	1,048,576	272,629,760
2	Reserved	273,678,336	16,777,216
3 (C:)	Basic	290,455,552	498,841,157,632
4	Recovery	499,131,613,184	1,027,604,480

Partition	Type	Offset (bytes)	Size (bytes)
5 (D:)	Basic	500,159,217,664	11,945,377,792

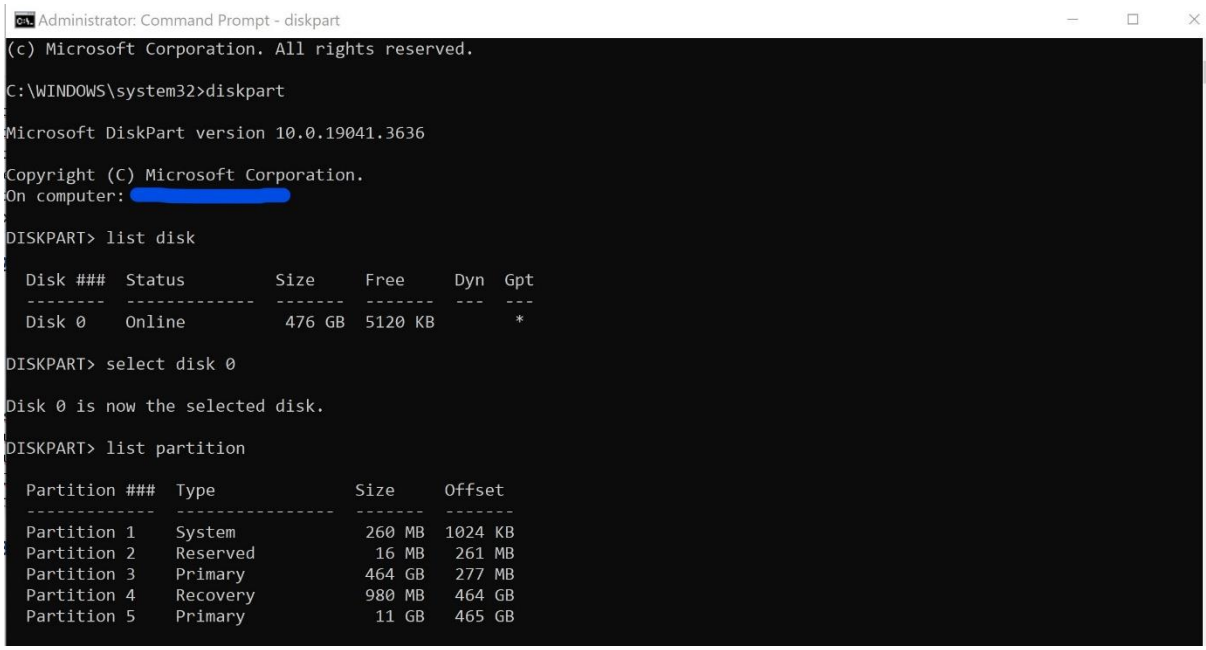


Screenshot 7b: Disk Partition in Powershell

D3 - DiskPart (listing only)

```
diskpart
list disk
select disk 0
list partition
exit
```

DiskPart showed the same five partitions. It also showed a star in the Gpt column, which means the disk is GPT. It shows sizes without decimals (for example 464 GB instead of 464.58 GB).



Screenshot 8: Disk Partition using DiskPart (listing only)

The three tools agree on the partition sizes. Only Disk Management does not show the 16 MB Reserved partition.

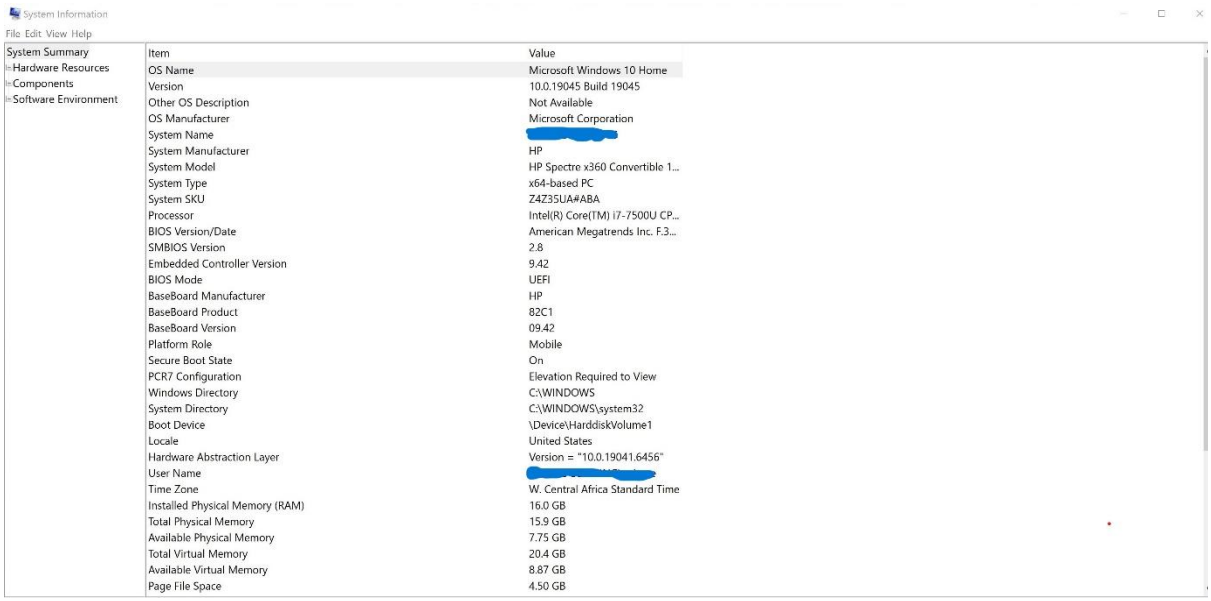
Part E - MBR, GPT and Boot Process Explanation

E1. How a PC starts

1. Power on: the hardware gets power and the processor starts.
2. POST (Power-On Self-Test): the computer checks that basic hardware works.
3. BIOS/UEFI firmware starts.
4. CMOS/firmware settings are read, such as boot order and Secure Boot.
5. The firmware finds the boot records (MBR/GPT or the EFI System Partition).
6. The boot loader starts and chooses the operating system.
7. The kernel (the core of the operating system) is loaded.
8. The operating system takes control, and logs and user activity start building up.

Why it matters in forensics: boot settings and boot records can show if a computer was changed or set up to start more than one operating system.

My system uses UEFI. System Information showed BIOS Mode: UEFI and Secure Boot State: On. This fits with my GPT disk and the 260 MB EFI System Partition.



Screenshot 9: Boot mode (UEFI)

E2. MBR and GPT

	MBR	**GPT**
Age	Older	Newer, used with UEFI
Disk size	About 2 TB	Very large disks (about 9.4 ZB)
Where partitions are listed	First sector (sector 0)	GPT header and entries, with a backup copy

	MBR	**GPT**
Forensic consideration	Damage to sector 0 can affect partition interpretation.	The backup GPT structures may help an investigator validate or recover partition information.

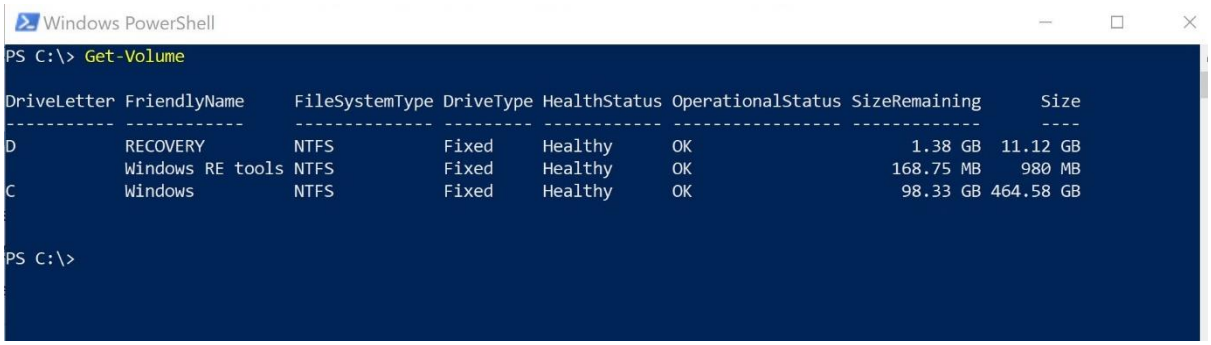
My disk uses GPT. Get-Disk showed Partition Style GPT, and DiskPart showed the star in the Gpt column.

Part F - File System Identification and Interpretation

Volume	File system	Where I saw it
C: (Windows)	NTFS	wmic logicaldisk, Get-Volume
D: (RECOVERY)	NTFS	wmic logicaldisk, Get-Volume
Windows RE tools (partition 4)	NTFS	Get-Volume



Screenshot 10a: File System Evidence



Screenshot 10b: File System Evidence

All the volumes I could identify use NTFS, the Windows file system. The EFI System Partition and the Reserved partition did not appear in Get-Volume, so I could not read their file system and I did not guess.

Why file systems matter: a file system is the way a partition stores files. An investigator must know it to read the data correctly. NTFS keeps records such as file times and a list of every file (the MFT), which help to build a timeline and recover files. Other common file systems are FAT32 (USB drives), ext4 (Linux) and HFS+ (older Macs). I only found NTFS on my laptop.

Part G - Final Practical Assignment

	Partition 1	**Partition 2**
Type	System (EFI System Partition)	Reserved
Start sector	2,048	534,528
End sector	534,527	567,295
Number of sectors	532,480	32,768
Size I calculated	0.2539 GB	0.0156 GB
Size Windows reports	260 MB	16 MB

My calculated sizes match what Windows reports. 0.2539 GB is 260 MB (0.2539 x 1024), and 0.0156 GB is 16 MB.

Why numbers can look different: I divide by 1024, which gives GB in the way computers count. Disk makers count 1 GB as 1,000,000,000 bytes, so the same size looks different. Tools also round differently. For example, partition 5 (D:) is exactly 11.125 GB, and Disk Management shows 11.13 GB, PowerShell shows 11.12 GB, and DiskPart shows 11 GB.

Final Summary and Conclusion

I found that my laptop is an HP Spectre x360 with Windows 10 Home, 16 GB of RAM and a 512 GB Samsung disk. It starts in UEFI mode with Secure Boot on, and the disk uses GPT with five partitions.

I calculated the size of partition 1 (0.2539 GB) and partition 2 (0.0156 GB) from their sectors, and the results match Windows (260 MB and 16 MB). I learned that size differences between tools are usually about units and rounding.

This matters in forensics because an investigator must record the system, disk layout and file systems accurately, using read-only methods, so the findings can be trusted.

Appendix

A. PowerShell command used to check my calculations

```
Get-Partition -DiskNumber 0 | Select-Object PartitionNumber,
  @{n='StartSector';e={$_.Offset/512}},
  @{n='Sectors';e={$_.Size/512}},
  @{n='EndSector';e={($_.Offset/512)+($_.Size/512)-1}},
  @{n='SizeGB';e={[math]::Round($_.Size/1GB,4)}} | Format-Table -AutoSize
```

Partition	Start sector	Sectors	End sector	Size (GB)
1	2,048	532,480	534,527	0.2539
2	534,528	32,768	567,295	0.0156
3	567,296	974,299,136	974,866,431	464.582
4	974,866,432	2,007,040	976,873,471	0.957
5	976,873,472	23,330,816	1,000,204,287	11.125

B. Screenshot checklist

No.	Screenshot required	Screenshot(s) in this report	Included?
1	System Information or systeminfo output	Screenshots 1a and 1b (Part A, A1)	Yes
2	Disk drive model and size output	Screenshot 2 (Part A, A2)	Yes
3	Logical volume / file system output	Screenshot 3 (Part A, A2)	Yes
4	Running process / PID output	Screenshot 4 (Part B, B1)	Yes
5	Disk Management, Get-Partition, DiskPart output	Screenshots 7a, 7b and 8 (Part D)	Yes
6	Partition 1 calculation evidence	Screenshot 6, row for partition 1 (Part C, C2)	Yes
7	Partition 2 calculation evidence	Screenshot 6, row for partition 2 (Part C, C2)	Yes
8	Boot mode evidence: BIOS/UEFI	Screenshot 9 (Part E, E1)	Yes
9	File system identification evidence	Screenshots 10a and 10b (Part F)	Yes
10	Final report summary / conclusion page	Conclusion section of this report	Yes